

Sketching Rational graphs

Basic skills

Q1. $y = \frac{1}{5x+2}$ asymptote at $x = -\frac{2}{5}$



Transformed $\frac{1}{x}$ graph

$$\frac{1}{x} \xrightarrow{\begin{bmatrix} -2 \\ 0 \end{bmatrix}} \frac{1}{x+2} \xrightarrow{\text{stretch SF: } \frac{1}{5} \text{ in } x\text{-dir}} \frac{1}{5x+2}$$

Q2. $y = \frac{2x-5}{5x+3}$ $\leftarrow N(x)$
 $\leftarrow D(x)$

a) $y=0$ when $N(x)=0$ so $2x-5=0$

so $x = \frac{5}{2} \leftarrow x\text{-intercept}$ $\uparrow$ y-intercept

b) At $x=0$ $y = \frac{2 \times 0 - 5}{5 \times 0 + 3} = -\frac{5}{3}$

c) Undefined when $D(x)=0$ so

at $5x+3=0 \Rightarrow x = -\frac{3}{5} \leftarrow \text{vertical asymptote}$

d) At either side of $x = -\frac{3}{5}$
we have $y \rightarrow \pm\infty$

To figure out which consider signs

As $x \rightarrow -\frac{3}{5}$ from below

$$y = \frac{-}{-} = + \quad \leftarrow \text{try } x = -1 \text{ for example}$$
$$y = \frac{-2}{-2} = \frac{2}{2}$$

As $x \rightarrow -\frac{3}{5}$ from above

$$y = \frac{-}{+} = - \quad \leftarrow \text{try } x = 0 \text{ for example}$$
$$y = \frac{-3}{3} = -\frac{3}{3}$$

$$e) \lim_{x \rightarrow \infty} \frac{2x-5}{5x+3} = \lim_{x \rightarrow \infty} \frac{2 - \frac{5}{x}}{5 + \frac{3}{x}} = \frac{2}{5}$$

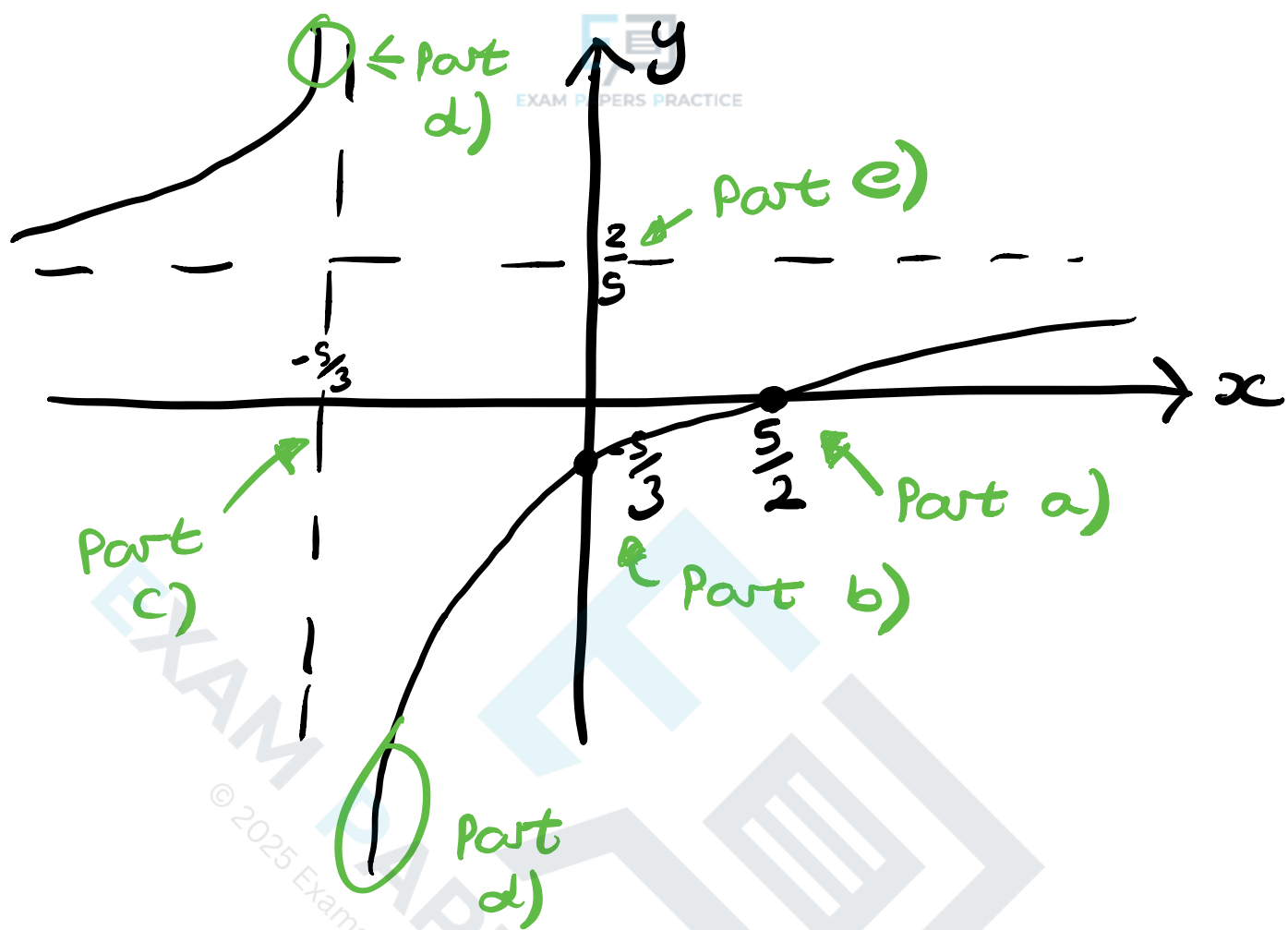
$$\lim_{x \rightarrow -\infty} \frac{2x-5}{5x+3} = \lim_{x \rightarrow -\infty} \frac{2 - \frac{5}{x}}{5 + \frac{3}{x}} = \frac{2}{5}$$

For direction we have

$$x \rightarrow \infty ; \quad \frac{2 - (+ve)}{5 + (+ve)} \quad \text{so from below}$$

$$x \rightarrow \infty ; \quad \frac{2 - (-ve)}{5 + (-ve)} \quad \text{so from above}$$

f)



Q3. $y = \frac{6x}{8-3x}$

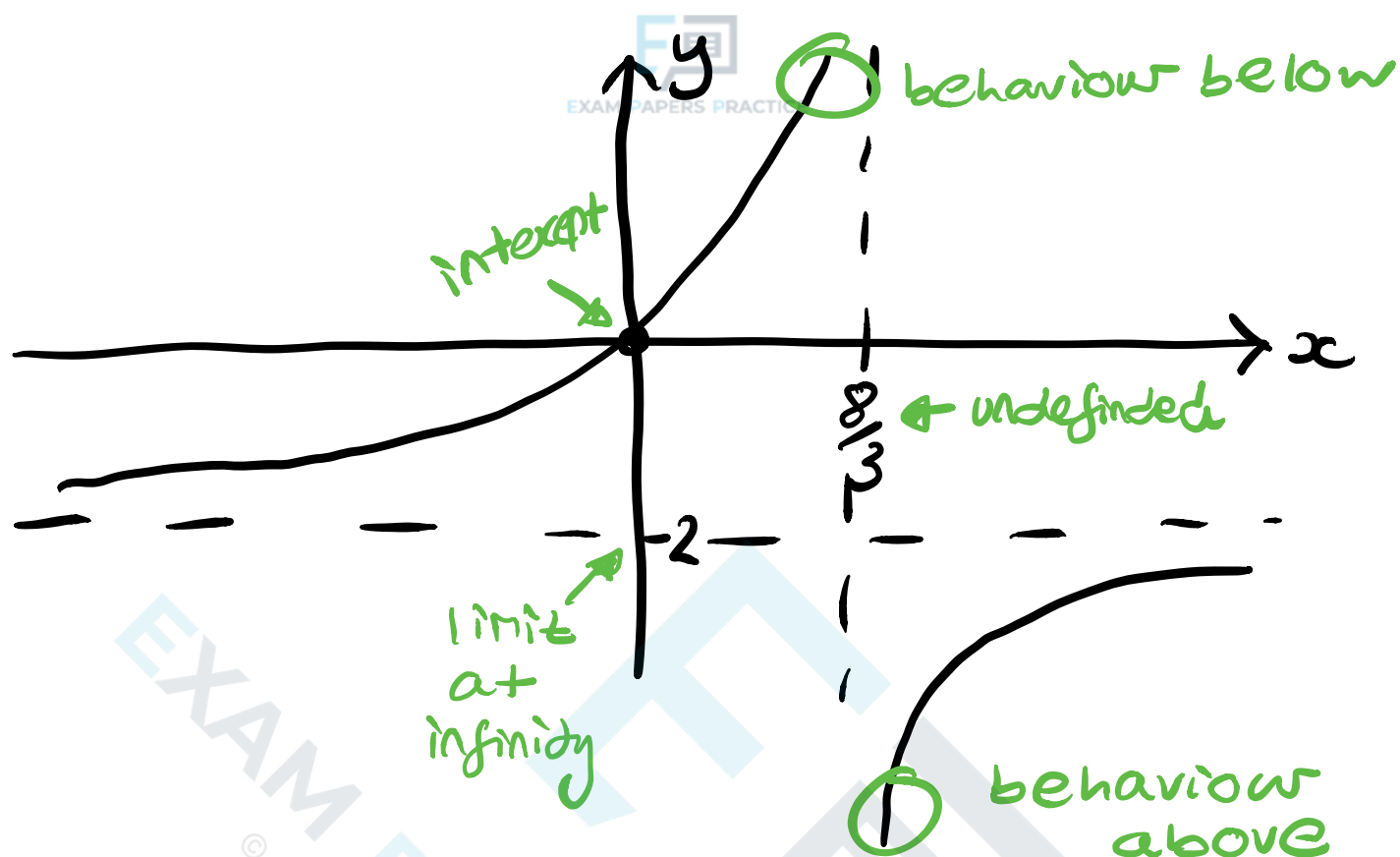
Intercepts: • At $x=0$ $y=0$ $(0,0)$
 • At $y=0$ $x=0$

undefined at $8-3x=0$ $x = \frac{8}{3}$

Above $\frac{8}{3}$ y is $\frac{+}{-} = -$ so $-\infty$

Below $\frac{8}{3}$ y is $\frac{+}{+} = +$ so $+\infty$

Limit as $x \rightarrow \pm\infty$ $y = \frac{6}{\frac{8}{x}-3} \rightarrow \frac{6}{-3} = -2$



Q4. $y = \frac{1}{(x-1)(x+2)}$

Intercepts: At $x = 0$ $y = \frac{1}{-2} = -\frac{1}{2}$

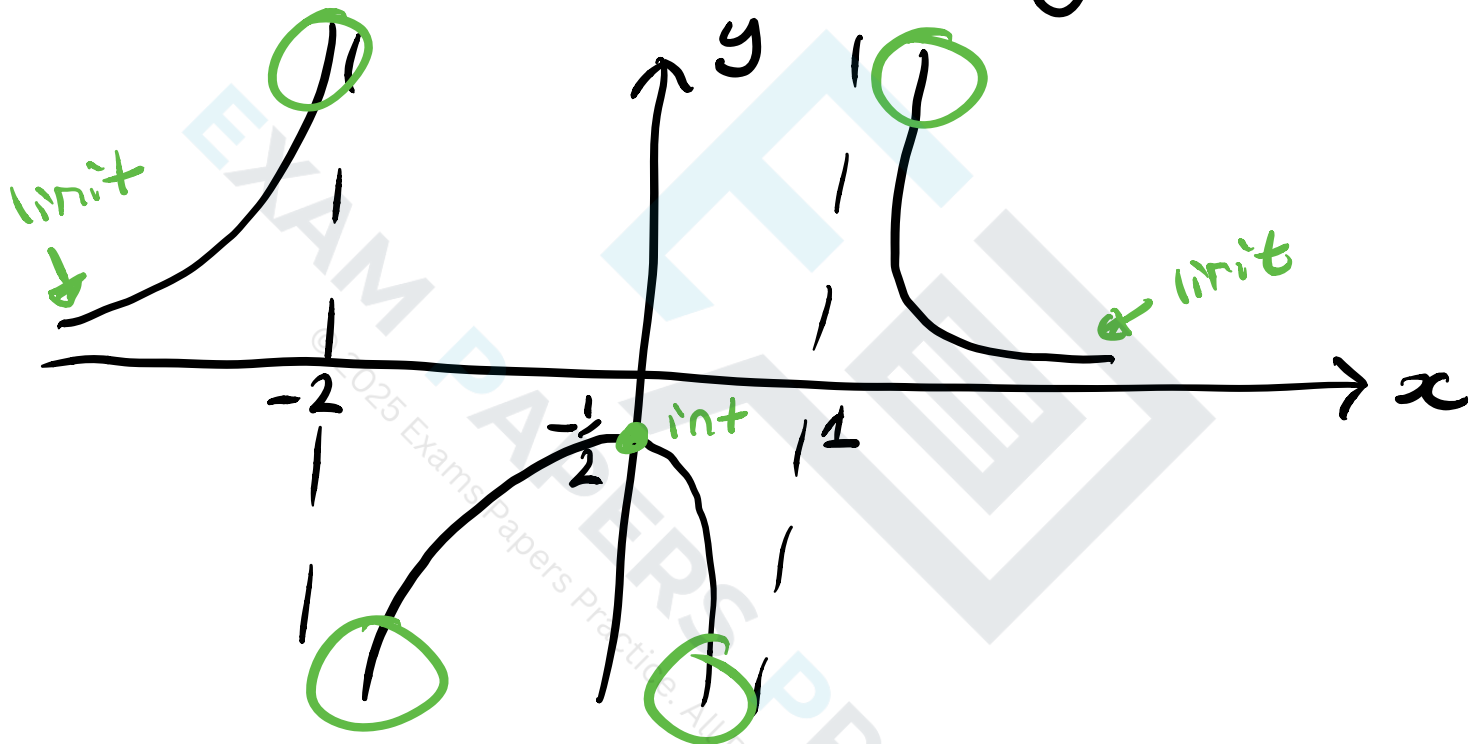
At $y = 0 \Rightarrow 1 = 0$ so no solutions!

undefined when $(x-1)(x+2) = 0$ so at $x = -2$ or 1 for behaviours of y at sides

For $x = -2$ $\begin{cases} \text{Below } y \text{ is } \frac{+}{-x-} = + \\ \text{Above } y \text{ is } \frac{+}{-x+} = - \end{cases}$

For $x = 1$ $\begin{cases} \text{Below } y \text{ is } \frac{+}{-x+} = - \\ \text{Above } y \text{ is } \frac{+}{+x+} = + \end{cases}$

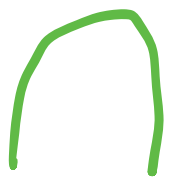
Limit as $x \rightarrow \pm\infty$ is $y \rightarrow 0$



note.

behaviours circled at sides of undefined asymptotes.

max of middle



section is not found but
can be by $\frac{dy}{dx} = 0$ (& $\frac{d^2y}{dx^2} < 0$)

Competency skills

Q5. $y = \frac{4x+3}{x^2+5x-24}$

a) Intercepts: At $x=0$ $y = \frac{3}{-24} = -\frac{1}{8}$

$$\begin{aligned} \text{At } y=0 &\Rightarrow 4x+3=0 \\ &\Rightarrow x = -\frac{3}{4} \end{aligned}$$

b) Undefined at $x^2+5x-24=0$
 $\Rightarrow (x+8)(x-3)=0 \Rightarrow x=-8, 3$

For behaviours

$$y = \frac{4x+3}{(x+8)(x-3)}$$

Around $x=-8$ { Below $y = \frac{-}{-x-} = -$

can try some
values eg. $x=-9$,
 $x=-7$

{ Above $y = \frac{-}{+x-} = +$

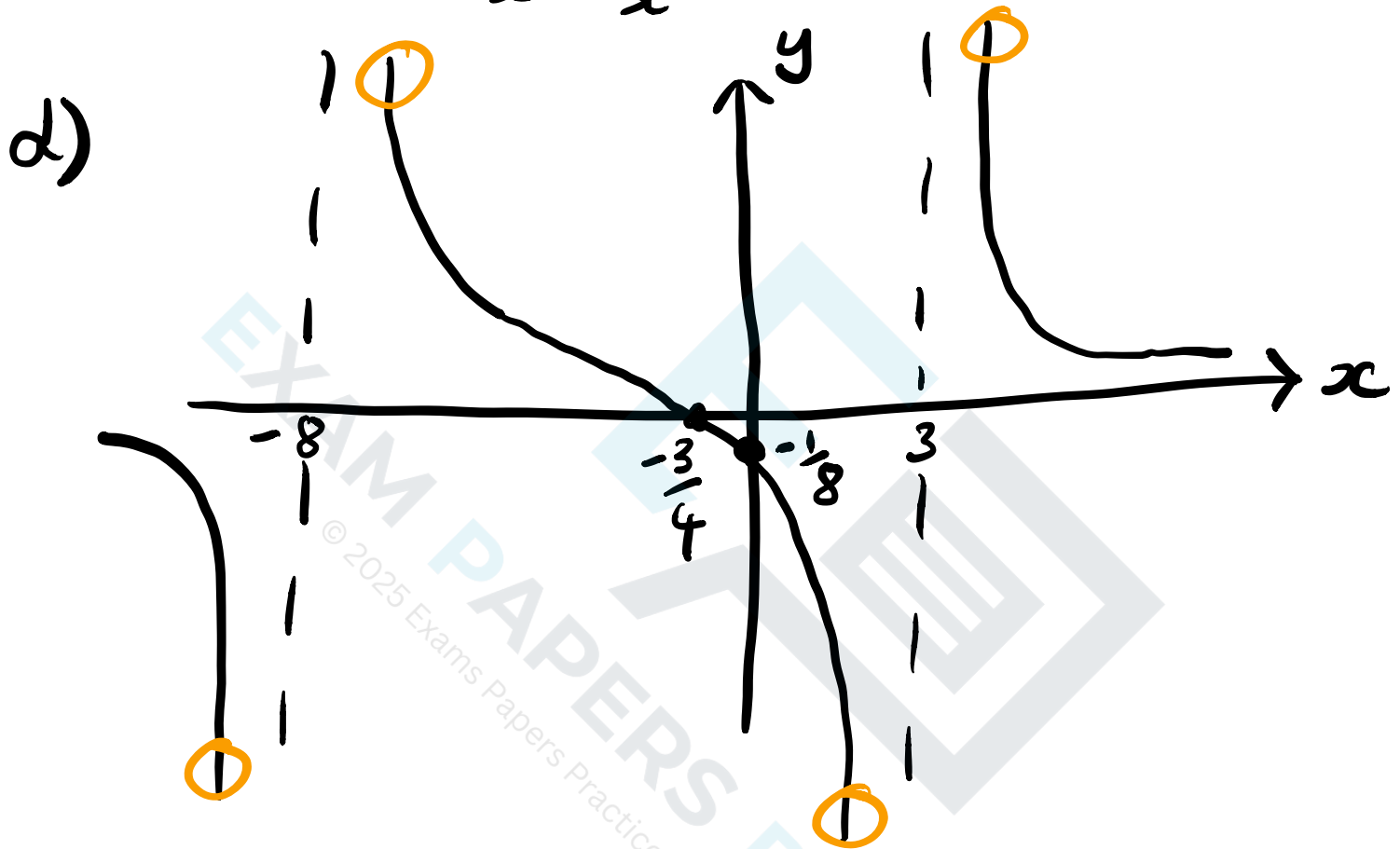
Around $x=3$ { Below $y = \frac{+}{+x-} = -$

eg. $x=2$
 $x=4$

{ Above $y = \frac{+}{+x+} = +$

or closer with calculator

c) $y = \frac{\frac{4}{x} + \frac{3}{x^2}}{1 + \frac{5}{x} - \frac{24}{x^2}} \rightarrow 0 \text{ as } x \rightarrow \pm \infty$



behaviour use

Q6. $y = \frac{x(x-1)}{(x-2)(x+3)}$

At $x=0$ $y=0$

At $y=0 \Rightarrow x(x-1)=0 \Rightarrow x=0, 1$

so $(0,0), (1,0)$

undefined at $(x-2)(x+3)=0$

$$x = -3, 2$$

Around $x = -3$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{-x-}{-x-} = + \\ \text{Above } y \text{ is } \frac{-x-}{-x+} = - \end{array} \right.$

Around $x = 2$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{+x+}{-x+} = - \\ \text{Above } y \text{ is } \frac{+x+}{+x+} = + \end{array} \right.$

Limit as $x \rightarrow \pm\infty$ is

$$y = \frac{1(1 - \frac{1}{x})}{(1 - \frac{2}{x})(1 + \frac{3}{x})} \rightarrow \frac{1 \times 1}{1 \times 1} = 1$$

For max/mins use quotient rule

on $y = \frac{x^2 - x}{x^2 + x - 6}$ $\leftarrow u \quad du = 2x - 1$
 $\leftarrow v \quad dv = 2x + 1$

to get $\frac{dy}{dx} = \frac{(x^2 + x - 6)(2x - 1) - (x^2 - x)(2x + 1)}{(x^2 + x - 6)^2}$



EXAM PAPERS PRACTICE

$$\text{For } \frac{dy}{dx} = 0 \Rightarrow (x^2 + x - 6)(2x - 1) - (x^2 - x)(2x + 1) = 0$$

$$\Rightarrow 2x^3 + x^2 - 13x + 6 - (2x^3 - x^2 - x) = 0$$

$$\Rightarrow 2x^2 - 12x + 6 = 0$$

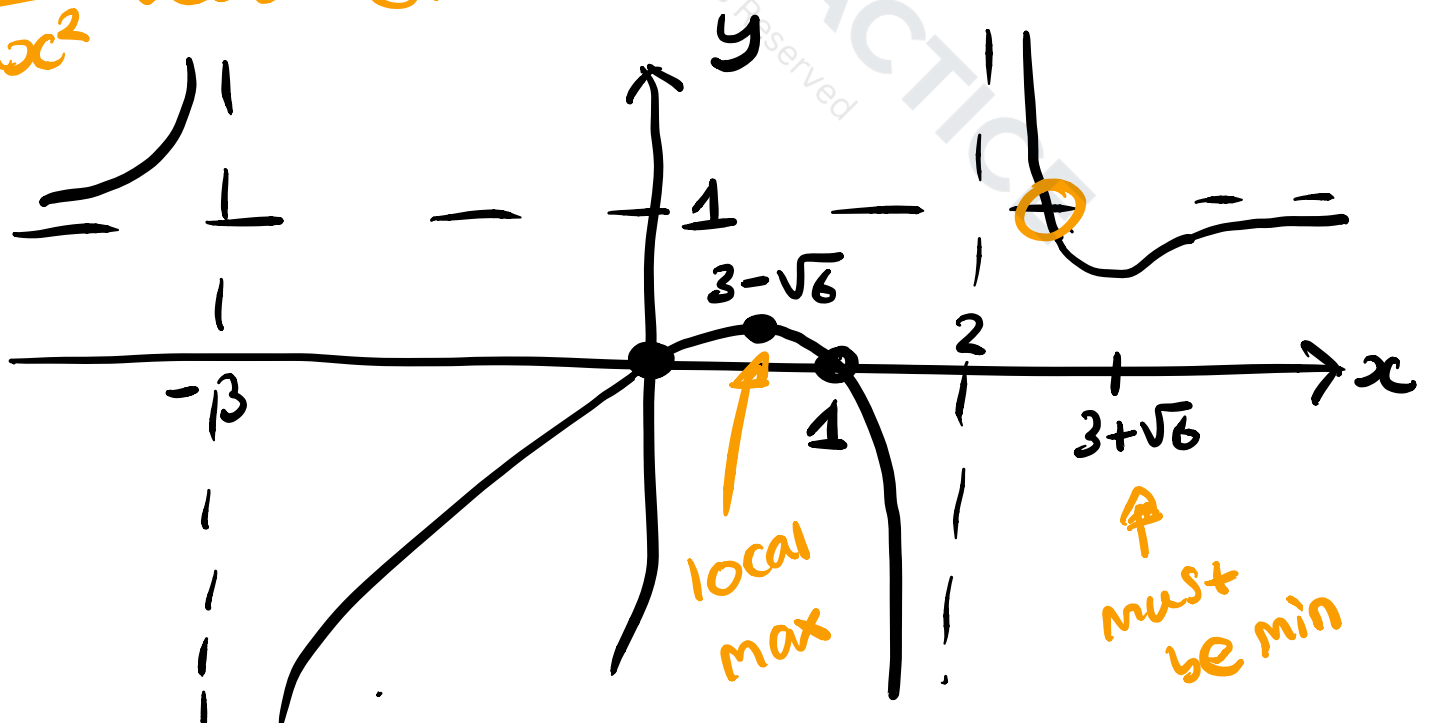
$$\Rightarrow x^2 - 6x + 3 = 0$$

$$\text{so } x = \frac{6 \pm \sqrt{36 - 12}}{2}$$

$$= \frac{6 \pm \sqrt{24}}{2} = 3 \pm \sqrt{6}$$

Could now use

$\frac{d^2y}{dx^2}$ test or draw



Looks like crosses limiting asymptote which is fine below 1

In fact As $x \rightarrow +\infty$ $y \rightarrow 1^-$

e.g At $x=100$ $y=0.98$

Could also check $y=1$ has a solution

Q7. To solve $\frac{x+5}{4x-2} \geq 2$ sketch

$y = \frac{x+5}{4x-2}$ and $y=2$ with intersection pts

For $y = \frac{x+5}{4x-2}$ Intercepts: $x=0 \Rightarrow y = -\frac{5}{2}$
 $y=0 \Rightarrow x+5=0 \Rightarrow x=-5$

Undefined at $4x-2=0$ $x=2$

Below $x=2$ y is $\frac{+}{-} = -$

Above $x=2$ y is $\frac{+}{+} = +$

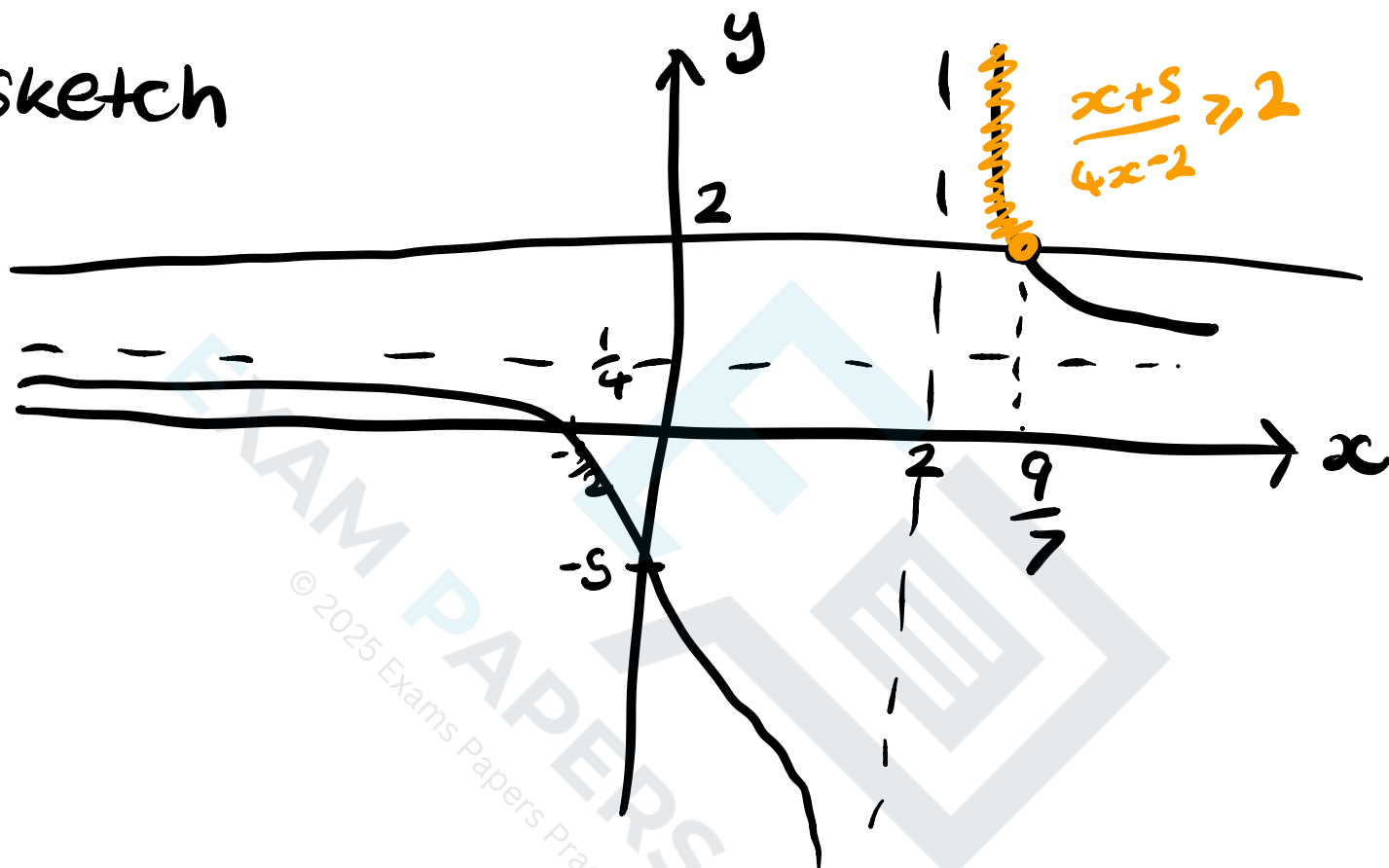
Limit as $x \rightarrow \pm\infty$ $y = \frac{1 + \frac{5}{x}}{4 - \frac{2}{x}} \rightarrow \frac{1}{4}$

For intersection with $y=2$

Solve

$$\frac{x+5}{4x-2} = 2 \Rightarrow x+5 = 8x-4$$
$$\Rightarrow 7x = 9 \Rightarrow x = \frac{9}{7}$$

Sketch



so

$2 < x \leq \frac{9}{7}$ is solution

Q 8. a) $y = \frac{x^2+x-6}{x^2-x-6} = \frac{(x+3)(x-2)}{(x-3)(x+2)}$

Intercepts: At $x=0$ $y = \frac{-6}{-6} = 1$

$$\text{At } y=0 \Rightarrow (x+3)(x-2)=0$$
$$\Rightarrow x = -3, 2$$

$(0, 1), (-3, 0), (2, 0)$

Undefined at $(x-3)(x+2)=0 \Rightarrow x=-2, 3$

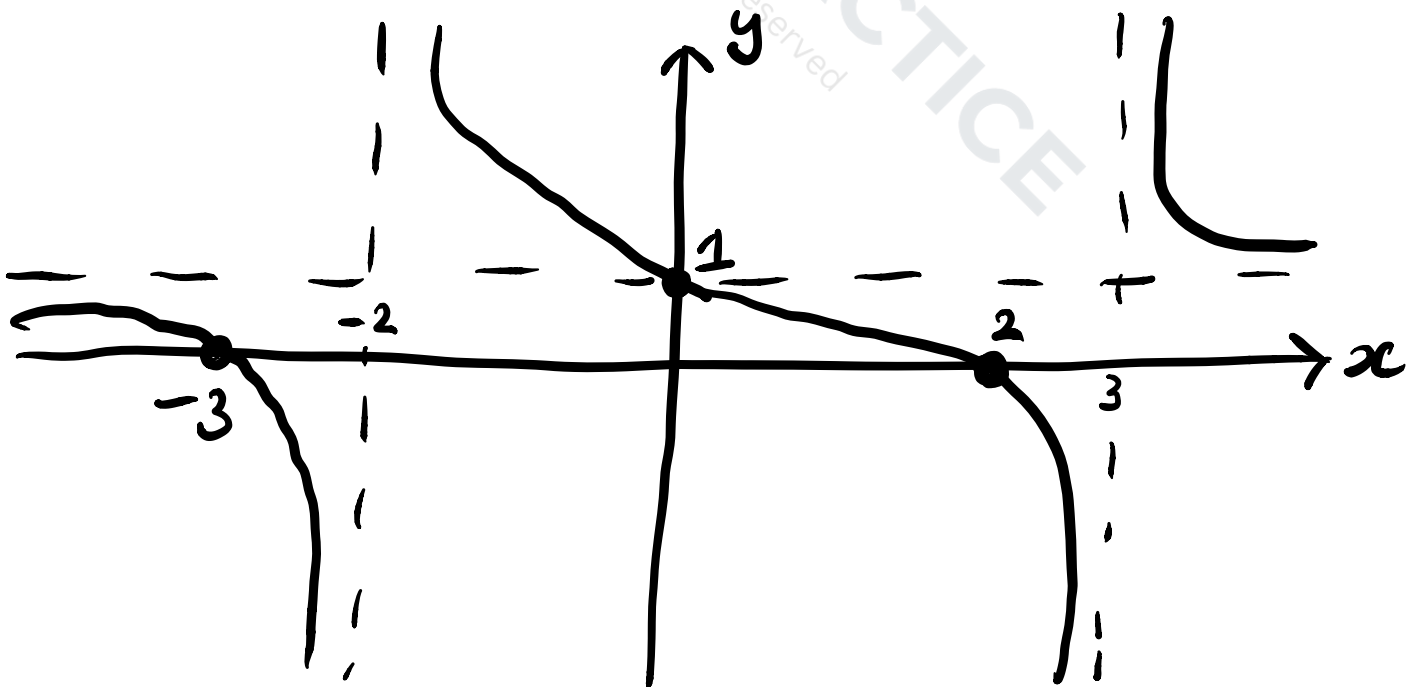
Behaviour

Around $x=-2$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{+x-}{-x-} = - \\ \text{Above } y \text{ is } \frac{+x-}{-x+} = + \end{array} \right.$

Around $x=3$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{+x+}{-x+} = - \\ \text{Above } y \text{ is } \frac{+x+}{+x+} = + \end{array} \right.$

Limit as $x \rightarrow \pm\infty$

$$y = \frac{(1+\frac{3}{x})(1-\frac{2}{x})}{(1-\frac{3}{x})(1+\frac{2}{x})} \rightarrow \frac{1 \times 1}{1 \times 1} = 1$$



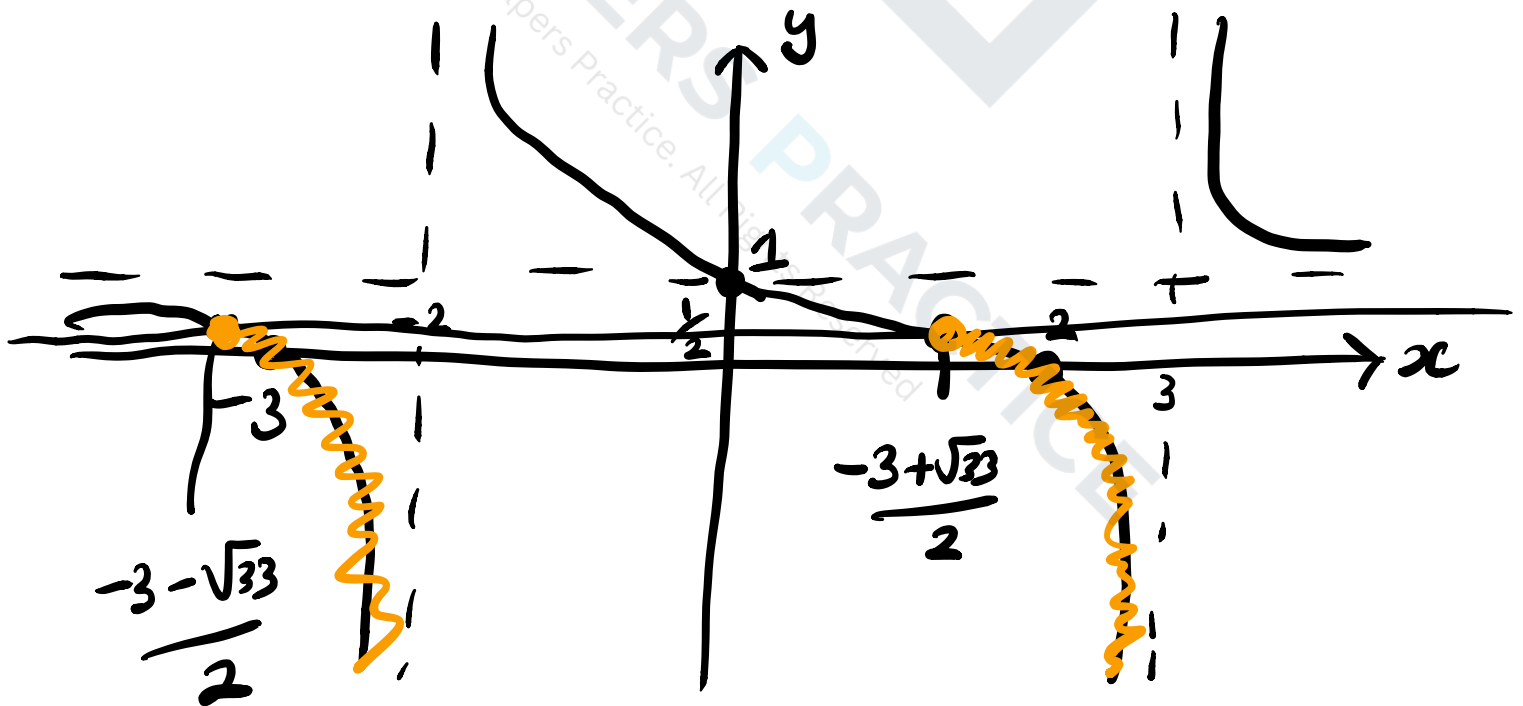
b) Find intersection points by

$$\frac{x^2 + x - 6}{x^2 - x - 6} = \frac{1}{2}$$

$$\Rightarrow 2x^2 + 2x - 12 = x^2 - x - 6$$

$$\Rightarrow x^2 + 3x - 6 = 0 \Rightarrow x = \frac{-3 \pm \sqrt{9+24}}{2}$$

$$x = \frac{-3 \pm \sqrt{33}}{2}$$



So

$$\frac{-3 - \sqrt{33}}{2} < x < -2, \quad \frac{-3 + \sqrt{33}}{2} < x < 3$$

Q9. $y = \frac{x^2 - 2x - 5}{x^3 + x^2 - 17x + 15}$ $\leftarrow N(x)$
 $\leftarrow D(x)$

a) Factorise with $x-1$ as factor as 1 is a root

$$(x-1)(x^2 + 2x - 15) = D(x)$$

solving $D(x) = 0$

$$\Rightarrow (x-1)(x^2 + 2x - 15) = 0$$

$$\Rightarrow (x-1)(x+5)(x-3) = 0$$

$\Rightarrow x = -5, 1, 3$ are vertical asymptotes

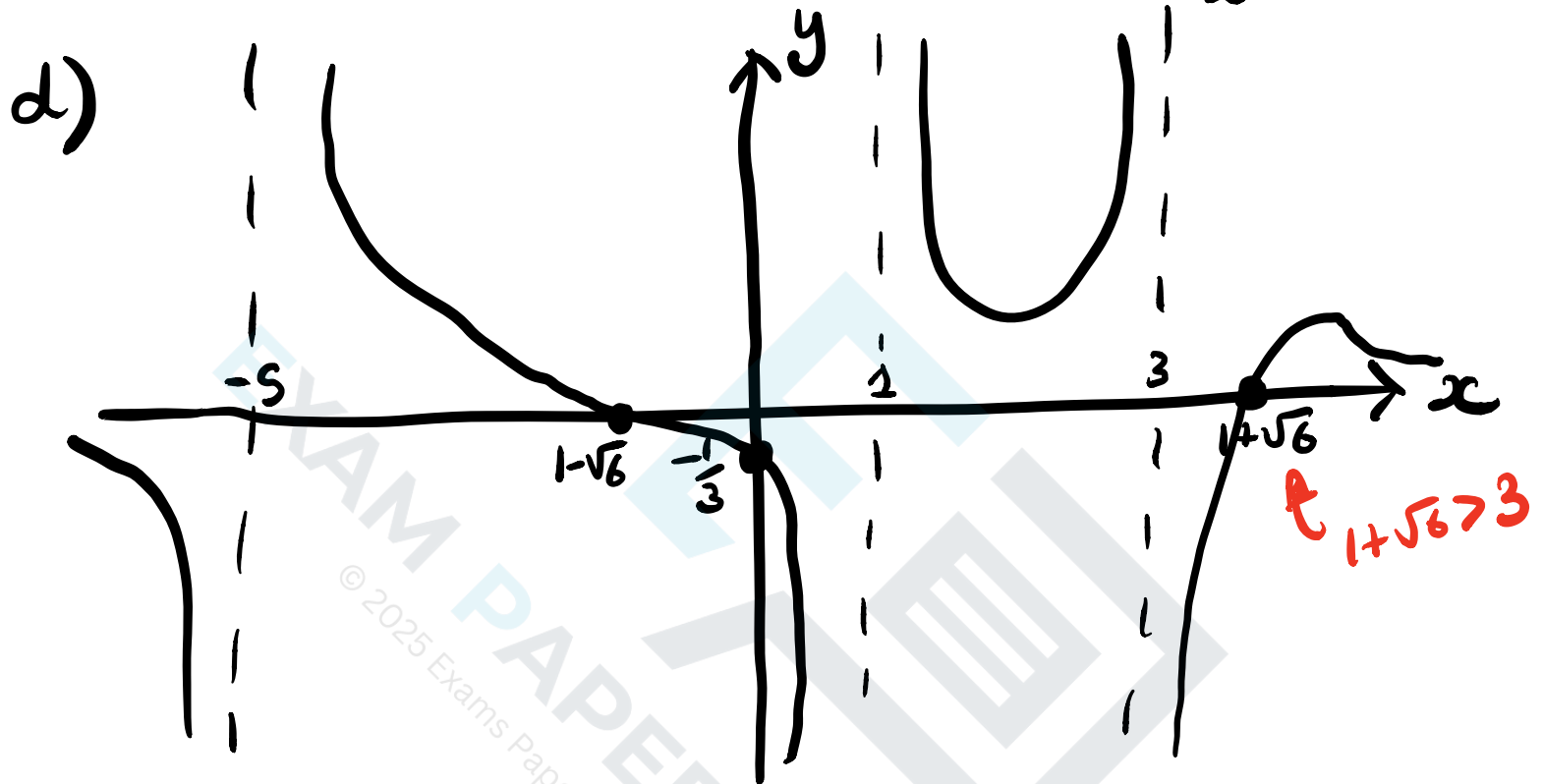
b) At $x = 0$ $y = \frac{-5}{15} = -\frac{1}{3}$

For $y = 0 \Rightarrow x^2 - 2x - 5 = 0$

$$\Rightarrow x = \frac{2 \pm \sqrt{24}}{2} = 1 \pm \sqrt{6}$$

$$(0, -\frac{1}{3}), (1 - \sqrt{6}, 0), (1 + \sqrt{6}, 0)$$

c) $\lim_{x \rightarrow \pm\infty} y = \lim_{x \rightarrow \pm\infty} \frac{\frac{1}{x} - \frac{2}{x^2} - \frac{5}{x^3}}{1 + \frac{1}{x} - \frac{17}{x^2} + \frac{15}{x^3}} = \frac{0}{1} = 0$



Need behaviours to make easier

Around $x = -5$ { Below $\frac{+}{-x-x-} = -$
Above $\frac{+}{+x-x-} = +$

Around $x = 1$ { Below $\frac{-}{+x-x-} = -$
Above $\frac{-}{+x+x-} = +$

Around $x = 3$ { Below $\frac{-}{+x+x-} = +$
Above $\frac{-}{+x+x+} = -$

Q10. $y = \frac{(x-5)(2x+3)}{(x-1)^2}$

Intercepts At $x=0$ $y = \frac{-5 \times 3}{(-1)^2} = -15$

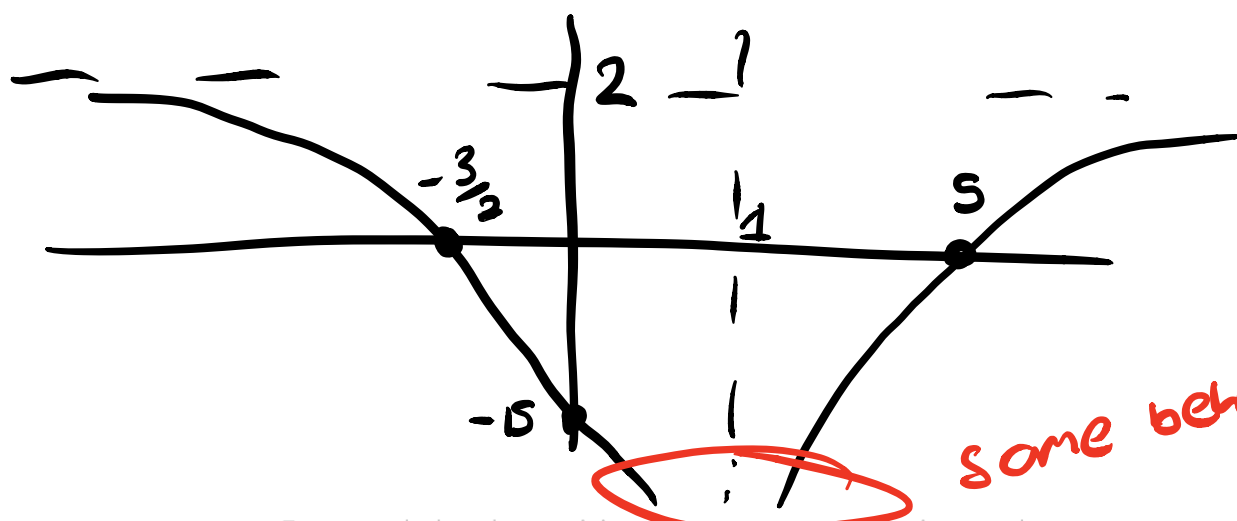
For $y=0 \Rightarrow (x-5)(2x+3)=0 \Rightarrow x = -\frac{3}{2}, 5$

Undefined at $x=1$

Around $x=1$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{-x+}{(-)^2} = - \\ \text{Above } y \text{ is } \frac{-x+}{(+)^2} = - \end{array} \right.$

$y = \frac{(1-\frac{5}{x})(2+\frac{3}{x})}{(1-\frac{1}{x})^2} \rightarrow \frac{1 \times 2}{1^2} = 2$
as $x \rightarrow \pm \infty$

Same
sign
behaviour



same behaviour

Q11. $y = \frac{x^n - a^n}{x - a}$

for $n=1$: $y = \frac{x-a}{x-a} \equiv 1$

$n=2$: $y = \frac{x^2 - a^2}{x-a} \equiv \frac{(x-a)(x+a)}{x-a}$

$\vdots$
 $\equiv x+a$

$x^n - a^n$ always has a factor of $x-a$ as it is the difference of the n^{th} power

So y is not a true rational as $D(x) = 1$ when simplified

Hence no vertical asymptotes